

3.3.4 Metals and alloys

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 7			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Unreactive metals, such as gold, are found in the Earth as the metal itself, but most metals are found as compounds that require chemical reactions to extract the metal. Metals less reactive than carbon can be produced by heating the metal compounds in the ore with carbon. Ores contain enough metal to make it economic to extract the metal. Large amounts of rock need to be quarried or mined to get metal ores.		5.4.1.3	4.8.2.1
We should recycle metals to save resources and limit environmental impacts. Students should be able to describe the social, economic and environmental impacts of mining ores and recycling metals.		5.10.2.2	4.8.2.9
Outcome 8			
Metals have giant structures of atoms with strong bonds between the atoms and so most metals have high melting points.	Suggested activity for TDA Compare the properties, such as conductivity or density, of some metals.	5.2.2.7	4.6.2.7
Metals are good conductors of electricity and thermal energy. Copper has properties that make it useful for electrical wiring and plumbing.	The properties of copper are limited to its ability to conduct electricity easily and the ease with which it can be worked.	5.2.2.8	4.6.2.7
Aluminium is a useful metal because of its low density and resistance to corrosion.	No knowledge of the extraction process of aluminium is required.		
Outcome 9			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Most metals in everyday use are alloys. Pure iron, gold and aluminium are too soft for many uses and so are mixed with small amounts of other elements to make alloys, which are harder for everyday use.	Knowledge of the composition of specific alloys is not required.	5.2.2.7	4.6.2.7
Most iron is converted into steels. Steels are alloys since they are mixtures of iron with carbon and other metals.	Suggested activity for TDA Investigate the hardness of different alloys or steels.		